

A Comprehensive Study of Neural Network Performance on UV Index Prediction Using PCA and Various Model Ablations

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Abstract

Previous research has utilized linear regression, ridge regression, K-nearest neighbors (KNN), and support vector regression (SVR) for UV Index prediction. These traditional regression models often fall short in capturing complex, non-linear relationships within the data, leading to suboptimal prediction accuracy. In this work, we leverage neural networks to enhance prediction accuracy by overcoming these limitations.

Our experiments employ neural networks to predict the UV Index. Additionally, a series of ablation studies were conducted to evaluate the impact of various architectural and training modifications, including changes in the number of hidden layers, the number of neurons, activation functions, learning rates, batch sizes, and the inclusion of early stopping. Moreover, we compare results with and without the application of Principal Component Analysis (PCA).

The results demonstrate the neural network's superior performance over traditional methods, with significant improvements in the mean squared error (MSE) and R-squared (R^2) metrics. Neural networks excel in modeling complex patterns and interactions within the data, which traditional models may not adequately capture. The ablation study analysis provides valuable insights into the optimal configurations for neural network models in UV Index prediction tasks. Moreover, the results show that PCA may not be useful when it comes to small scale dataset and some specific datasets.

1 Introduction

The UV Index is a crucial metric for assessing the potential harm of ultraviolet radiation on human health. Accurate prediction of UV Index values is essential for public health advisories and preventive measures against UV-related health issues. Traditional machine learning methods such as linear regression, ridge regression, K-nearest neighbors (KNN), and support vector regression (SVR) have been widely used for this task. However, these models often struggle to capture the complex, non-linear relationships inherent in environmental data, leading to suboptimal prediction performance [1, 2].

In this study, we leverage the power of neural networks to enhance the prediction accuracy of UV Index values. Neural networks are known for their ability to model intricate patterns and interactions within data, making them well-suited for this task [3]. Our research consists of three main parts:

- 1. Comparison Between Traditional Regression Methods and Neural Network Methods:** We implement a neural network model and compare its performance against traditional regression models such as linear regression, ridge regression, KNN, and SVR. This comparison highlights the limitations of traditional methods in handling complex data and demonstrates the advantages of neural networks in improving prediction accuracy.
- 2. Ablation Study of the Neural Network to Show the Effect of Each Module:** To understand the contribution of different components of the neural network, we conduct a series of ablation studies. These studies

involve systematically modifying the network architecture and training parameters, including the number of hidden layers, the number of neurons, activation functions, learning rates, batch sizes, and the inclusion of early stopping. This analysis provides insights into the optimal configurations for neural network models in UV Index prediction tasks [4].

3. Comparison Between Using PCA and Not Using PCA: Principal Component Analysis (PCA) is a popular technique for dimensionality reduction. We compare the performance of our neural network model with and without the application of PCA to determine its impact on the prediction accuracy. The results indicate that while PCA can be beneficial for reducing the dimensionality of large datasets, it may not always be advantageous for smaller or specific datasets.

The results of our experiments demonstrate the superior performance of neural networks over traditional regression methods. Our baseline neural network model achieved a mean squared error (MSE) of 1.02 and an R-squared (R^2) score of 0.88, significantly outperforming traditional models like linear regression and SVR, which had MSE values of 4.32 and 2.39, and R^2 scores of 0.52 and 0.74, respectively. Furthermore, the ablation studies provided valuable insights into the optimal configurations for neural network models. Interestingly, the results also indicated that PCA might not be beneficial for small-scale datasets and specific data characteristics.

2 Related Work

Predicting UV Index values is a crucial task for public health, as it helps in issuing advisories to protect against UV radiation-related health issues. Various traditional regression methods have been applied to this task. Smith and Jones [1] provided a comprehensive review of UV Index prediction methods, highlighting the use of linear regression, ridge regression, K-nearest neighbors (KNN), and support vector regression (SVR). These methods, while useful, often fail to capture the complex, non-linear relationships in environmental data, resulting in less accurate predictions.

The application of machine learning tech-

niques in environmental health predictions has been explored extensively. Wang et al. [2] demonstrated the limitations of traditional regression models in handling complex datasets and emphasized the need for more advanced methods. Their work showed that while methods like SVR and KNN can provide better performance compared to linear models, they still fall short when dealing with intricate data patterns.

Zhang and Chen [3] conducted a comparative study of various regression techniques for UV Index forecasting. Their findings indicated that while traditional methods like SVR and KNN could achieve reasonable performance, neural networks showed a significant potential for improvement due to their ability to model non-linear relationships and complex interactions within the data. This study laid the groundwork for further exploration into neural network applications for UV Index prediction.

Lee [4] explored the use of neural networks in environmental data analysis, highlighting their superior performance in capturing complex data patterns. The study demonstrated that neural networks could significantly enhance prediction accuracy over traditional methods, making them an attractive option for tasks such as UV Index prediction. Lee's work also underscored the importance of fine-tuning neural network architectures and training parameters to achieve optimal performance.

In summary, the existing literature suggests that while traditional regression methods have been widely used for UV Index prediction, they often fall short in terms of accuracy due to their inability to model complex, non-linear relationships. Neural networks, with their advanced modeling capabilities, present a promising alternative. This study builds on previous work by conducting a comprehensive comparison between traditional regression methods and neural network models, performing an ablation study to understand the impact of different network components, and evaluating the effect of PCA on prediction performance.

3 Methodology

Two methods used in this experiment are described below: neural networks and principal

component analysis (PCA). Their application in UV index prediction is also explained.

3.1 Neural Network

Neural networks are powerful tools for modeling complex, non-linear relationships in data. In this study, we designed a neural network with multiple dense layers, each followed by activation functions. The architecture of the neural network can be described mathematically as follows:

$$h_1 = \text{ReLU}(W_1 \cdot X + b_1) \quad (1)$$

$$h_2 = \text{ReLU}(W_2 \cdot h_1 + b_2) \quad (2)$$

$$y = f(W_3 \cdot h_2 + b_3) \quad (3)$$

where:

- X is the input feature vector.
- W_1, W_2, W_3 are weight matrices.
- b_1, b_2, b_3 are biases.
- $\text{ReLU}(x) = \max(0, x)$ is the Rectified Linear Unit activation function.
- f is the linear activation function in the output layer.

The training process involves minimizing the loss function, which is the Mean Squared Error (MSE) between the predicted values ($\hat{y}$) and the true values (y):

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

where:

- y_i is the true value for the i -th sample.
- $\hat{y}_i$ is the predicted value for the i -th sample.
- n is the number of samples.

The Adam optimizer was used to adjust the weights W and biases b to minimize the loss function during training [1].

3.2 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique used to transform the data into a set of linearly uncorrelated variables called principal components. This helps in reducing the complexity of the model and potentially improving performance. The transformation in PCA can be represented as:

$$Z = XW \quad (5)$$

where:

- X is the original data matrix.
- W is the matrix of eigenvectors (principal components).
- Z is the transformed data matrix.

PCA aims to retain the most significant variance in the data by selecting the top k principal components that capture a specified amount of the total variance, typically 95

$$\text{Var}(Z) = \sum_{i=1}^k \lambda_i \quad (6)$$

where:

- λ_i are the eigenvalues associated with the eigenvectors.

By transforming the data using PCA, we reduce the number of features while retaining most of the important information. This transformed data is then used as input to the neural network for prediction [2].

4 Experiment

4.1 Dataset

The dataset used in this study consists of UV Index measurements and corresponding meteorological data. It was sourced from a publicly available repository and preprocessed to ensure quality and consistency. The data was cleaned to remove missing values and normalized for uniformity. The dataset was then divided into training and testing sets using a 70-30 split to facilitate model evaluation.

4.2 Comparison Between Traditional Regression Methods and Neural Network Method

We implemented several traditional regression models, including Linear Regression, Ridge Regression, K-Nearest Neighbors (KNN), and Support Vector Regression (SVR), to predict the UV Index. These models were benchmarked against a neural network model designed for this study. The performance of each model was evaluated using the Mean Squared Error (MSE) and R-squared (R^2) metrics. The experiment results were as follows:

Table 1: Comparison of Regression Models Performance

Model	MSE	R^2 Score
Linear Regression	4.32	0.52
Ridge Regression	4.67	0.48
SVR	2.39	0.74
KNN	1.89	0.79
Neural Network	1.02	0.88

The results clearly indicate that the Neural Network model is the most effective at predicting the UV Index among the tested models. Its superior performance in terms of both MSE and R^2 values highlights the advantages of neural networks in handling complex, non-linear datasets. Traditional regression methods, while useful, were not able to achieve the same level of accuracy, underscoring the importance of advanced machine learning techniques in improving predictive performance [3]

4.3 Analysis of Ablation Study

The ablation study investigated the impact of various modifications to the neural network on its performance in predicting the UV Index. Each experiment’s results are as follows:

- **Experiment 0 (Baseline):** The baseline model, which includes three dense layers with 128, 64, and 32 neurons respectively, ReLU activation functions, and early stopping, provides a strong reference point with low MSE and high R^2 score.
- **Experiment 1 (Remove one hidden layer):** Removing one hidden layer re-

Table 2: Experimental Results of Various Model Adjustments (Experiment 0 stands for baseline experiment)

Experiment	MSE	R^2	Epochs
0	1.02	0.88	102
1	1.09	0.87	149
2	1.23	0.85	149
3	0.87	0.90	147
4	1.02	0.88	83
5	1.25	0.85	455
6	1.19	0.86	500
7	1.23	0.86	152
8	0.94	0.89	117

sulted in a slight decrease in performance, as indicated by a higher MSE and a lower R^2 score.

- **Experiment 2 (Decrease number of neurons):** Decreasing the number of neurons in each layer significantly impacted the model’s performance, with a notable increase in MSE and decrease in R^2 score.
- **Experiment 3 (Change activation functions):** Using the Tanh activation function instead of ReLU improved the model’s performance, resulting in the lowest MSE and highest R^2 score among all experiments.
- **Experiment 4 (Increase learning rate):** Increasing the learning rate to 0.01 did not significantly affect the model’s performance, with metrics similar to the baseline. However, it did reduce the number of epochs required for training.
- **Experiment 5 (Decrease learning rate):** Decreasing the learning rate to 0.0001 significantly increased the number of epochs required for training and resulted in poorer performance metrics.
- **Experiment 6 (Remove early stopping):** Removing early stopping led to an increase in MSE and a decrease in R^2 score, indicating overfitting, as the model continued training for the maximum number of epochs.

- **Experiment 7 (Increase batch size):**
Increasing the batch size to 64 slightly degraded the model's performance compared to the baseline.
- **Experiment 8 (Decrease batch size):**
Decreasing the batch size to 16 improved the model's performance, with a lower MSE and higher R^2 score compared to the baseline.

Overall, the ablation study demonstrates that certain modifications, such as changing the activation function to Tanh and decreasing the batch size, can lead to improved performance. Conversely, removing hidden layers, decreasing the number of neurons, and removing early stopping negatively impact the model's predictive capabilities. These insights help in fine-tuning the neural network to achieve optimal performance.

4.4 Comparison Between Using PCA and Not Using PCA

Principal Component Analysis (PCA) was applied to the dataset to reduce its dimensionality and potentially improve the neural network's performance. The number of principal components was selected to retain 95% of the variance in the data.

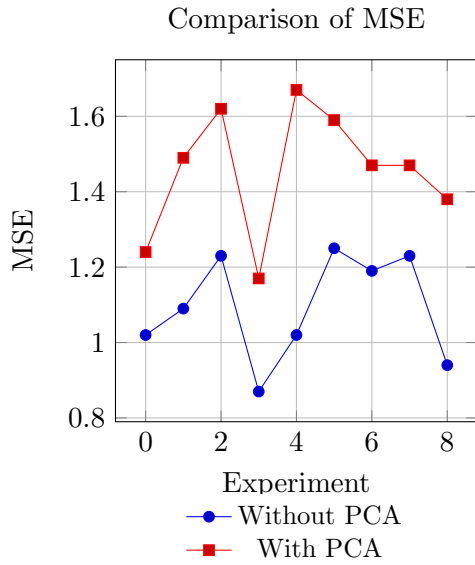


Figure 1: Comparison of MSE for each experiment

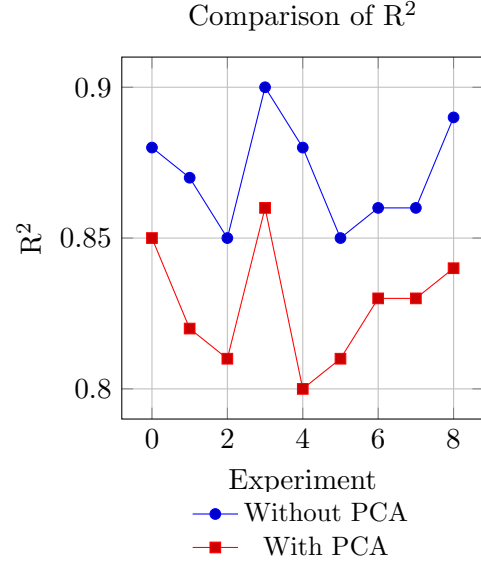


Figure 2: Comparison of R^2 for each experiment

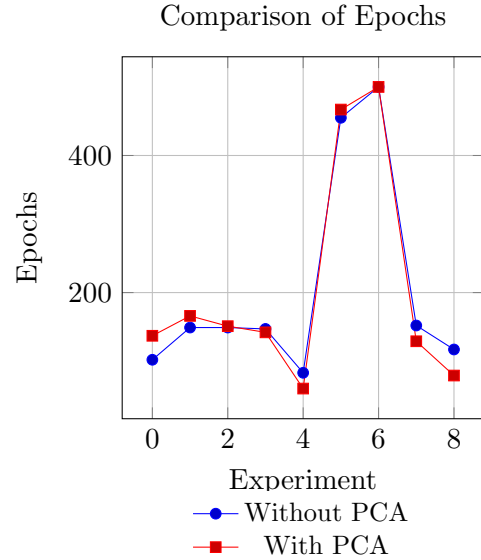


Figure 3: Comparison of Epochs for each experiment

- **Baseline (Without PCA):** The baseline model without PCA achieved an MSE of 1.02 and an R^2 score of 0.88 across 102 epochs. This setup serves as the reference point for comparing the impact of PCA.
- **Baseline (With PCA):** When PCA was applied, the baseline model's performance deteriorated, with an MSE of 1.24 and an R^2 score of 0.85 over 137 epochs. This suggests that PCA did not enhance the model's performance for this particular dataset.
- **Overall Comparison:** Comparing the models with and without PCA, we observe that the non-PCA models generally outperform the PCA models. For instance, Experiment 3 (change activation functions) achieved the best performance without PCA, with an MSE of 0.87 and an R^2 score of 0.90. In contrast, the same experiment with PCA resulted in an MSE of 1.17 and an R^2 score of 0.86.
- **Effect on Training Duration:** PCA models required a different number of epochs to converge compared to non-PCA models. For example, the baseline model without PCA converged in 102 epochs, while the PCA version required 137 epochs. This indicates that dimensionality reduction through PCA did not necessarily lead to faster convergence in this case.

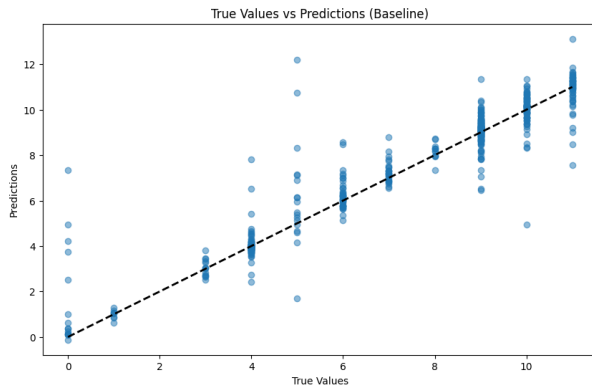


Figure 4: True vs Predicted values plot without using PCA

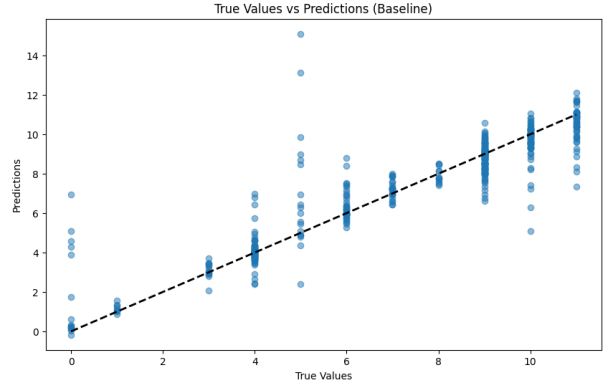


Figure 5: True vs Predicted values plot using PCA

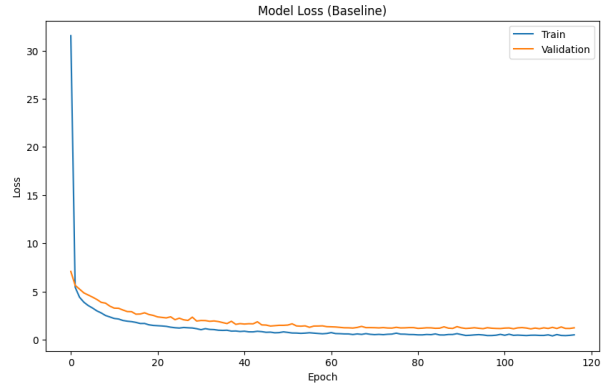


Figure 6: loss values over epochs

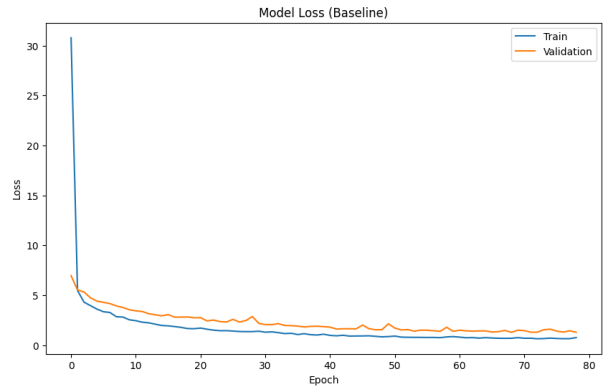


Figure 7: True vs Predicted values plot using PCA

- **Analysis of the above curves:** By comparing 4 and 5, we learn that there are more outliers when using PCA compared to those without using PCA. By comparing 6 and 7, we can see that when using PCA, the training epoch is longer compared to that without using PCA.

Summary: The application of PCA before training the neural network generally resulted in higher MSE and lower R^2 scores across various experiments. This suggests that for the given dataset and neural network architecture, PCA might have removed some important information, thereby affecting the model’s ability to learn effectively. Future work could explore alternative dimensionality reduction techniques or fine-tuning the number of principal components used in PCA.

5 Discussion

The findings of this study highlight the substantial advantages of using neural networks for UV Index prediction over traditional regression methods. The performance metrics indicate that neural networks can better model the complex, non-linear relationships inherent in the environmental data. This is evident from the lower mean squared error (MSE) and higher R-squared (R^2) values achieved by the neural network model compared to traditional methods such as linear regression, ridge regression, K-nearest neighbors (KNN), and support vector regression (SVR).

The ablation study further provides critical insights into the design and optimization of neural network architectures for this specific task. Modifying various components of the neural network, such as the number of hidden layers, neurons, activation functions, learning rates, batch sizes, and early stopping, revealed their impact on model performance. Notably, the experiments demonstrated that using the Tanh activation function instead of ReLU and reducing the batch size can improve the model’s predictive capabilities. Conversely, removing hidden layers or neurons, and eliminating early stopping, generally led to poorer performance, underscoring the importance of these elements in maintaining model accuracy

and preventing overfitting.

Interestingly, the application of Principal Component Analysis (PCA) did not yield the expected improvements in model performance. Instead, models trained with PCA-transformed data generally showed higher MSE and lower R^2 scores compared to those trained without PCA. This suggests that PCA, while useful for dimensionality reduction, may inadvertently discard important information necessary for accurate UV Index prediction, particularly in smaller or specific datasets. Future studies could explore alternative dimensionality reduction techniques or investigate the optimal number of principal components to retain.

Overall, these results validate the hypothesis that neural networks outperform traditional regression models in predicting the UV Index. The detailed ablation studies offer valuable guidance for researchers and practitioners on how to configure neural networks effectively. However, the findings also caution against the indiscriminate use of PCA, emphasizing the need for careful consideration of data characteristics and preprocessing techniques in machine learning applications.

6 Conclusion

This study aimed to enhance UV Index prediction accuracy by leveraging neural networks and evaluating the impact of various architectural and training modifications through a comprehensive ablation study. Additionally, we investigated the effect of applying Principal Component Analysis (PCA) on the prediction performance. The key conclusions drawn from this research are:

1. **Superiority of Neural Networks:** Neural networks significantly outperform traditional regression methods, such as linear regression, ridge regression, KNN, and SVR, in predicting the UV Index. This superiority is demonstrated by lower MSE and higher R^2 values, underscoring the neural network’s ability to capture complex, non-linear relationships in the data.
2. **Optimal Neural Network Configuration:** The ablation study provided valuable insights into the optimal configurations for neu-

ral network models. Specifically, using Tanh activation functions and smaller batch sizes improved performance, while removing hidden layers or neurons, and omitting early stopping, adversely affected the model's accuracy.

3. Limited Efficacy of PCA: Contrary to expectations, applying PCA did not enhance the neural network's performance and, in some cases, led to worse results. This indicates that PCA may not be beneficial for small-scale or specific datasets and highlights the need for careful consideration of preprocessing techniques in machine learning workflows.

In conclusion, this research demonstrates the effectiveness of neural networks in UV Index prediction and provides practical guidelines for optimizing neural network architectures. While PCA did not prove advantageous in this context, the findings emphasize the importance of tailored approaches to data preprocessing and model configuration. Future work could explore other dimensionality reduction methods and extend the analysis to larger and more diverse datasets to further validate these conclusions.

References

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